

Multivariate statistics

Lecture 12:

Grouping and clustering of multivariate observations

Literature and Problems: See course schedule

Agenda (preliminary):

Introduction

- Problem statement
- Purpose and applications

Definition of distances between observations

- Continuous variables: Euclidean, m-norm (standardization)
- Binary (indicator) variables: #matches / #mismatches similarity metrics
- (distance between variables: correlation coefficients)

Definition of distances between clusters: linkage metrics

- **DEMO: inter-cluster linkage methods**

Motivation for need of search algorithms for definition of clusters

- # of possible clusters for n observations

Hierarchical clustering using distance matrix, D, and linkage metric

- **SLIDE: Hierarchical clustering algorithm and dendrograms**
- **DEMO: Hierarchical clustering**

Non-hierarchical clustering using data matrix, X (or Z)

- **SLIDE: "K-means" clustering algorithm**
- **DEMO: K-means clustering**

Clustering using Gaussian Mixture Models (GMM)

- Introduction to GMM
- **DEMO: Gaussian Mixture Model**
- **SLIDE: GMM clustering algorithm: estimation of parameters, calculation of posteriors, assignment of observations to clusters using MAP**
- **DEMO: GMM clustering**

Multidimensional scaling and visualization of clusters for high-dimensional data

- **SLIDE: Classical Metric Multidimensional scaling (short intro)**
- **DEMO: Multidimensional scaling (example)**